DAY13-PORTS-PROTOCOLS

THEORY (answer all 20 questions in your own words):

1. What is a network port?
   The port is a logical communiation endpoint used by services and network applications.

2. How many port numbers are available conceptually?
   Port number range 0 – 65535

3. What is a network protocol?
   A network protocol is a set of rules that define device communication over network.
   Eg: TCP, IP, HTTP etc…

4. What is TCP?
   TCP – Transmission control protocol
   Connection oreinted, reliable delivery, ordered, user acknowledgement.
   Web pages – SSH, email protocols, HTTP/HTTPS etc…

5. What is UDP?
   UDP – User Datagram Protocol
   Connectionless, low oriented, No built in delivery guarantee.
   Used for some real time/ query traffic.

6. Explain two differences between TCP and UDP.
   - TCP is connection-oriented and establishes a connection before sending data
   - UDP is connectionless.

   - TCP provides reliable delivery and error checking.
   - UDP is faster but does not guarantee delivery.

7. What is a listening port?
   A listening ports means a local service is waiting for incoming connections on a socket.
   Eg: TCP/22 – SSH service
   TCP/443 – HTTP service.

8. What is an ephemeral/client port?
   An ephemeral/client port is a temporary port assigned by a client device for a network connection. It allows the device to communicate with a server.

9. What is a socket?
   A socket is a communication endpoint combining IP address + protocol + port.

10. Which port is commonly used by SSH: TCP/22

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11. Which port is commonly used by HTTP: TCP/80

12. Which port is commonly used by HTTPS: TCP/433

13. Which port is used by DNS: UDP/53

14. Which UDP ports are commonly used by DHCP client/server: UDP 67/68

15. What is TCP port 445 commonly associated with: SMB [Server Message Block].

16. What is port 3389 commonly associated with: RDP [Remote Desktop Protocol].

17. What does ss -tuln display: Listening Ports

18. Does an open port automatically mean a vulnerability? Explain.
    No. An open port indicates a listening service. But can check for security risks.

19. Why do firewalls use protocol and port information?
    To allow or block network traffic based on rules.

20. Why are ports and protocols important in cybersecurity?
    They help identify network services, monitor traffic, and detect unauthorised access.

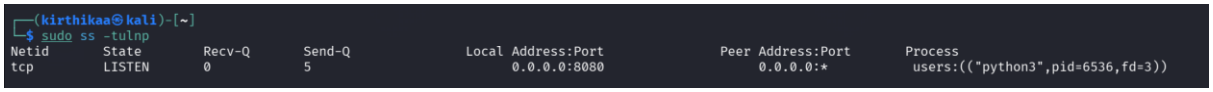
DAY 13 PRACTICAL LAB — Local Ports & Services Investigation

AUTHORIZED SYSTEMS ONLY: your own Kali/Linux VM, your own computer, or the Jayvik Labs lab. NO scanning external hosts today.

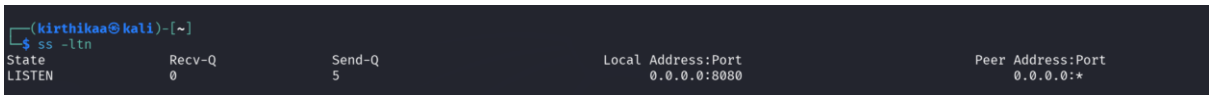
TASK 1 — ss -tuln: identify TCP listeners, UDP listeners, port numbers and local addresses. Explain -t -u -l -n.



TASK 2 — Optional if allowed: sudo ss -tulnp (-p can show process info; sudo only if you want it). Identify ONE listening service if available. NO services? Totally acceptable — do NOT enable unnecessary services.



TASK 3 — ss -ltn: record up to THREE TCP listening ports (Port / Address / State). No specific ports required.



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TASK 4 — Match: 22, 53, 80, 443, 445, 3389 → SSH, DNS, HTTP, HTTPS, SMB, RDP.

Port Number	Port
22	SSH - TCP
53	DNS – TCP + UDP
80	HTTP - TCP
443	HTTPS - TCP
445	SMB - TCP
3389	RDP- TCP/UDP

TASK 5 — TCP or UDP? SSH→TCP · HTTP→TCP · HTTPS→TCP (traditional HTTP/1.1/2 usage) · DHCP→UDP · DNS→UDP AND TCP · NTP→UDP. Some modern protocols add transports — that's normal.

Protocol	Transport
SSH	TCP
HTTP	TCP
HTTPS	TCP (traditional HTTP/1.1 and HTTP/2)
DHCP	UDP
DNS	UDP and TCP
NTP	UDP

TASK 6 — Client/server example: Client 192.168.1.20:54213 → Server 203.0.113.20:443. Which is the server port (443)? Which is the ephemeral client port (54213)? What service is 443 commonly? (HTTPS).

- Server port 443 is HTTPS in TCP connection
- 443 is server communication with encryption
- Ephemeral client port : 54213

TASK 7 — curl -I https://example.com: note HTTP status, the destination service (HTTPS) and its default port (443). A plain standard request — NOT scanning.



TASK 8 — Scenario: a web server should expose only 80 and 443, but an admin finds a listener on 23. Which legacy protocol uses TCP 23? (Telnet) Why investigate? (It sends sessions without modern encryption — unnecessary exposure / possible misconfiguration.) No attack steps here — investigation mindset only.

Protocol : Telnet uses TCP 23

Investigation: sends data, including login credentials such as username and password without encryption, creating a security risk and necessary exposure.

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